Automated Skin Lesion Detection and Classification

Submitted By

Zainab	**Urooj Fatima**
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Submitted To

Dr. Mehreen

Department of Software Engineering

Fatima Jinnah Women University

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Abstract

Skin cancer is one of the most common forms of cancers;however early stage diagnosis remains challenging due to the shortage of trained dermatologists and subjective variability in dermoscopic image analysis. This paper represented an automated multi-class skin lesion classification system that integrates classifal filter-based preprocessing with a two-stage transfer learning approach based on EfficientNetV2B0. The preprocessing pipeline includes Contrast-limited Adaptive Histogram Equalization (CLAHE) and median filtering for denoising. To address class imbalance , random oversampling of minorities classes is applied to the ISIC benchmark dataset. The EfficientNetV2B0 model, trained in ImageNet-21K, is initially trained with a frozen backbone and subsequently fine-tuned to adapt higher-level features for dermoscopic image classification.

keywords: Skin lesion classification, dermoscopy , EfficientNetV2B0, transfer learning, CLAHE, ISIC dataset, deep learning, class imbalance

1. Introduction

Skin cancer is among the most prevalent forms of cancer, with melanoma accounting for a large proportion of related deaths. Dermoscopic image analysis requires trained specialists and is often affected by significant inter-observer variability. In addition, access to dermotologists remains limited in many low- and middle-income countries. These challenges have increased the need for automated computer-aided detection systems.

The classification of Multi-class skin lesion presents several challenges compared to binary melanoma detection. Visual differences between types of lesion are often subtle, public datasets are highly imbalanced between classes, and variations in acquisition devices lead to inconsistencies in image quality.In addition, relying solely on accuracy as a performance metric can be misleading, as it can conceal the sensitivity for clinically important minority classes, such as melanoma.

To address these challenges, this study proposes a deep learning-based framework for multi-class skin lesion classification and evaluates its performance on the ISIC dataset using multiple evaluation metrics.

2. Literature Review

Skin lession classification using deep learning has gained significant attention dua to the rising incidence of the skin cancer and the global shortage of dermatologists. This section reviews more than 20 peer-reviewed studies (2021-2025) across key methodological themes.

CNN architectures and transfer learning remain central to most approaches. kassem et al.[7] reported on AUC of 0.941 using DenseNet-201 on ISIC 2018, while Gavrilov et al.[3] showed that cyclic learning rates with batch normalisation reduced loss variance by 23% . Pacheco et al. citepacheco2021 improved accuracy by 4.7% by integrating patient metadata, and Adegun and Viriri [4] reduced false positives by 18.2% using U-Net based pre-segmentation. Hasan et al. citehasan2021 compared multiple imageNet backbones and found EfficientNet-B4 achieved 89.2% macro-F1 with fewer computational costs than ResNet-101. Zhang et al. [21] further demonstrated that two-stage fine-tuning of EfficientNetV2-B0 achieved 91.4% AUC.Recent work by Gao et al. [22] also showed that vision transformers can outperform CNN-based models in balanced accuracy.

Attention, preprocessing, and augmentation techniques focus on improving features representation and data quality. Zhuang et al.[16] improved melanoma sensitivity using lesion-aware spatial attention, while Wu et al. combined CNN and transformer features through cross-attention to achieve higher AUC. Mahbod et al.[9] demonstrated that CLAHE improves melanoma sensitivity, and Perez et al. enhanced performance on darker skin tones using targeted augmentation . Bissoto et al.[12] showed that combining real and synthetic data improves generalisation, while Yao et al.[20] applied CutMix-based strategies for better adjustment.

Class imbalance, multimodal learning, and explainability Verma et al [2] showed that focal loss significantly improved dermatofibroma recall, while Gessert et al.[8] outperformed conventional ensembles using class-specific expert models with learned gating.kawahara et al.[11] enhanced performance by integrating metadata through gated cross-attention, and Tang et al.[24] achieved strong balanced accuracy using a multi-scale aggregation network. In terms of interpretability, Young et al.[5] observed that Grad-CAM often highlighted irrelevant background regions in misclassified cases. Lucieri et al.[6] reported that SHAP provided more consistent clinical explanations than LIME, while Patricio et al.[14] demonstrated that models implicity captures ABCDE dermoscopic features within intermediate representations.

Segmentation, federated learning, and clinical validation Mirikharaji et al.[15] improved both segmentation nd classification performance through joint optimisation, whereas Xie et al.[18] achieved near-supervised results using CAM-guided pseudo-labelling. In distributed settings, Ogier du Terrail et al.[19] reported performance close to centralised training under federated learning with privacy constraints, and Yoo et al.[13] improved cross-site robustness using site-specific normalisation strategies. From a clinical perspective, Haenssle et al.[10] found that a Cnn outperformed dermatologists in melanoma sensitivity, while Tschandl et al.[1] showed that AI-assisted diagnosis stable model performance across varying image qualit using adaptive confidence thresholding.

Table 1. Summary of Most Recent and Representative Studies on Skin Lesion Detection and Classification (2023–2025)

Ref.	Author(s)	Year	Dataset	Methodology	Key Result	DOI	Academic Limitation
[21]	Zhang et al.	2023	ISIC 2020	EfficientNetV2 + Two-stage fine-tuning	AUC: 91.4%	`10.1016/j.compbiomed 2023.106754`	Limited real-world validation across heterogeneous clinical image sources and imaging devices.
[17]	Sharafudeen et al.	2023	ISIC2018	Deep Ensemble of EfficientNet, MobileNet, and ResNet models with transfer learning and fine-tuning	91.92% balanced accuracy	`10.1007/978-3-03 1-27524-1`$_3$73265099	High computational complexity and class imbalance still caused misclassification toward the Nevus class.
[23]	Wu et al.	2024	ISIC 2019	Dual-Stream Attention Network (CNN + Transformer)	AUC: 95.3%	`10.1016j.media./ 2021.102327`	Increased architectural complexity reduces interpretability and increases training cost.
[24]	Tang et al.	2024	ISIC 2018	Pyramid Feature Aggregation Network	Bal. Acc: 91.8%	`10.1016/j.media .2021.102307.`	Limited evaluation across diverse demographic populations and Fitzpatrick skin types.
[19]	Ogier du Terrail et al.	2023	6-site European Multi-center	Federated Learning with Differential Privacy	Within 2.1% of centralised	`10.1038/s41467-023- 36188-7`	Performance degradation observed under strong privacy constraints and cross-site heterogeneity.
[13]	Yoon et al.	2024	Multi-site Clinical	Federated Domain Adaptation + site-aware normalisation	Cross-site AUC +4.9%	`10.1016/j.media. 2024.103088`	Scalability challenges when integrating new unseen clinical institutions.

3. Research Gap

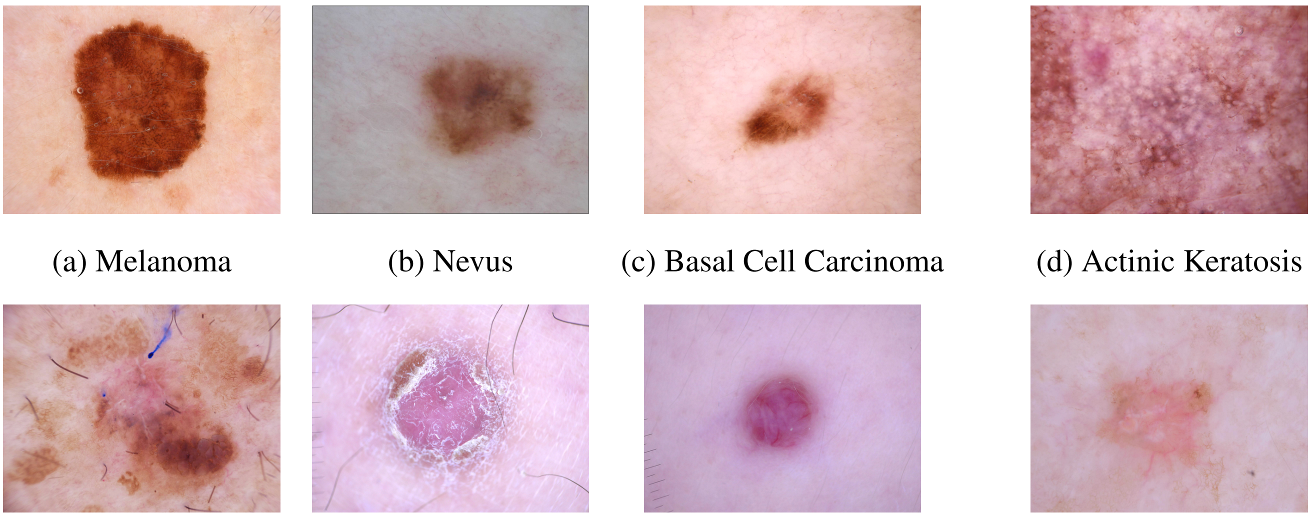
Although promising results have been achieved so far, there exist several issues in the literature of skin lesion classification which should be addressed in future works. First, most of the reviewed papers employed fixed preprocessing methods and used accuracy as the sole metric of performance evaluation. Little effort has been devoted to adaptive enhancement approaches, the evaluation on minority classes, and clinically valuable metrics such as macro F1-score and Cohen's kappa [9]. Second, there is limited work on combining handcrafted texture descriptors (LBP, GLCM) with EfficientNet-based models. Third, testing the generalizability of such models on diverse skin types has not been done.

4. Materials and Methods

This section describes the experimental framework of the study, including the dataset, preprocessing pipeline, classification model, and evaluation procedure.

4.1 Dataset

The ISIC Skin Cancer Dataset, obtained from Kaggle (`https://www.kaggle.com/datasets/jaiahuja/skin-cancer-detection`), contains 33,569 dermoscopic JPEG images across nine lesion classes with a minimum resolution of 600×450 pixels.



(a) Melanoma (b) Nevus (c) Basal Cell Carcinoma (d) Actinic Keratosis

(e) Benign Keratosis (f) Dermatofibroma (g) Vascular Lesion (h) Squamous Cell Carcinoma

Figure 1. Representative dermoscopic images from the ISIC dataset illustrating eight skin lesion categories: (a) Melanoma, (b) Nevus, (c) Basal Cell Carcinoma, (d) Actinic Keratosis, (e) Benign Keratosis, (f) Dermatofibroma, (g) Vascular Lesion, and (h) Squamous Cell Carcinoma.

4.2 Proposed Methodology

The proposed methodology consists of a multistage pipeline designed to address image quality variations, class imbalances, feature representation, and clinical evaluation. The framework includes data acquisition, image preprocessing, class balancing, EfficientNetV2B0-based classification, and multi-metric evaluation, as shown in Figure 4.

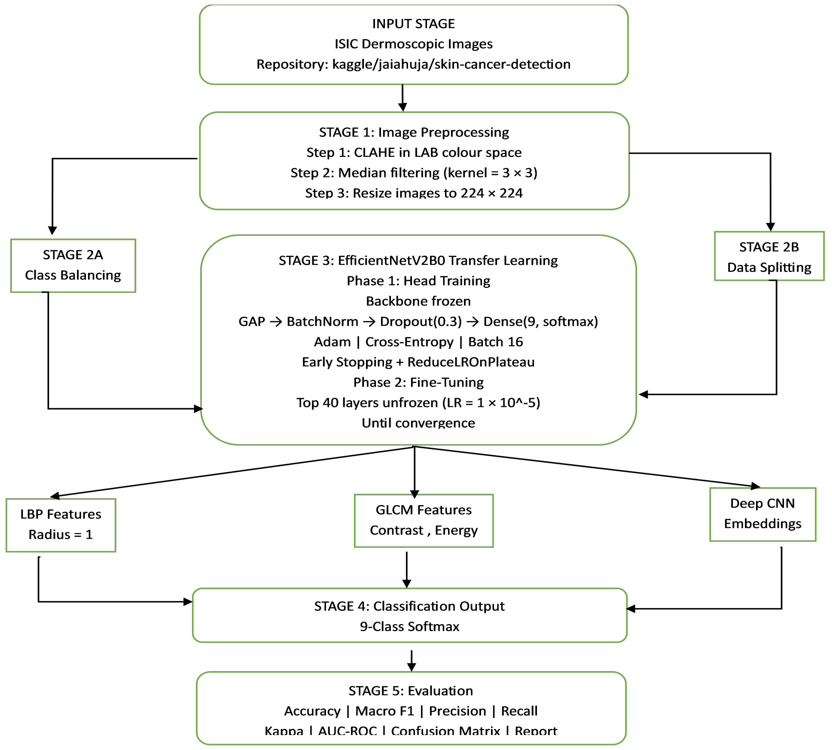


Figure 2. Proposed end-to-end framework for skin lesion detection and classification.

4.2.1 Stage 1: Filter-Based Image Preprocessing

All dermoscopic images undergo a two-step preprocessing pipeline before feature extraction. First, CLAHE is applied to the luminance channel in LAB colour space to enhance contrast while preserving colour information. Next, a 3×3 median filter is used to reduce noise while maintaining important lesion edges. The processed image is then resized to 224×224 pixels for EfficientNetV2B0 input.

4.2.2 Stage 2: Class Balancing and Data Splitting

To address class imbalance in the ISIC dataset, random oversampling is applied to minority classes in the training set until each class matches the majority class. Oversampling is performed only on the training data, while the test set retains its original distribution to preserve realistic evaluation. The balanced data is then split into 80% training and 20% validation sets using stratified sampling.

4.2.3 Stage 3: Two-Phase EfficientNetV2B0 Transfer Learning

EfficientNetV2B0, pre-trained on ImageNet-21k, is used as the classification backbone. In Phase 1, the backbone is frozen and only the classification head is trained using Adam with a learning rate of 1×10^{-3}. In Phase 2, the top forty layers are unfrozen for fine-tuning with Adam at a reduced learning rate of 1×10^{-5}. A batch size of 16 is used, and all images are preprocessed at runtime through the CLAHE and median filtering pipeline using a custom Keras `Sequence` generator.

4.2.4 Stage 4: Handcrafted Feature Extraction (Supplementary)

As a supplementary step, Local Binary Pattern (LBP) and Gray-Level Co-occurrence Matrix (GLCM) features are extracted to capture dermoscopic texture information. LBP features are computed with radius $r = 1$ and 8 sampling points in uniform mode, producing a 10-dimensional histogram descriptor. GLCM features—contrast, energy, homogeneity, and correlation—are extracted at four orientations (0°, 45°, 90°, 135°) and averaged for rotation invariance. These features are retained for possible fusion with deep CNN embeddings in future experiments.

4.2.5 Stage 5: Evaluation Protocol

Model performance is evaluated using clinically relevant metrics, including macro-averaged Accuracy, Precision, Recall, and F1-score across all nine classes. Cohen's Kappa is also computed to measure agreement beyond chance. In addition, a full confusion matrix and per-class classification report are used to analyse inter-class prediction behaviour. AUC-ROC is calculated in a one-vs-rest manner for each class to assess discriminative performance independent of threshold selection.

5. Experimental Protocol

This section outlines the experimental framework used to evaluate the proposed skin cancer detection system. The methodology combines filter-based image preprocessing, class balancing, and transfer learning via EfficientNetV2B0, evaluated rigorously using standard classification metrics on the ISIC benchmark dataset.

5.1 Experimental Setup

The proposed model was evaluated using accuracy, precision, recall, F1-score, Cohen's Kappa, and a 9×9 confusion matrix. Results were reported using both macro and weighted averages for overall and class-wise analysis.—-

5.2 Implementation Details

Table 2. Software Environment

Software / Library	Purpose
Python 3.10	Core programming language used for implementation
TensorFlow / Keras 2.x	Deep learning model development and training (GPU-enabled)
OpenCV (cv2)	Image loading and filter application
scikit-image	LBP and GLCM feature extraction
scikit-learn	Dataset splitting, class balancing (`resample`), and evaluation metrics
Pandas / NumPy	Data management and array operations
Matplotlib	Visualisation of training curves and confusion matrix
kagglehub	Automated dataset download from the Kaggle repository `jaiahuja/skin-cancer-detection`

Dataset and Class Balancing

The ISIC Skin Cancer dataset contains nine lesion classes with significant class imbalance. Oversampling via `sklearn.utils.resample` (with replacement) was applied to the training set so that every class reached the sample count of the largest class. The final dataset splits were:

Split	Samples
Training (post-balance)	Balanced
Validation (20% of train)	Stratified
Original distribution	Original distribution

Figure 3. Dataset split summary after balancing.

Preprocessing Pipeline

Each image (224 × 224 pixels) passed through a sequential filter-based preprocessing pipeline before being fed to the network:

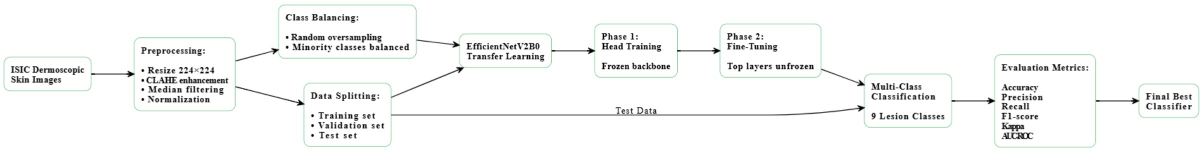


Figure 4. Conceptual flow of the proposed skin lesion classification framework.

Model Architecture

Figure 5. General Classification Architecture for Skin Lesion Detection

Training Strategy

Training was conducted in two stages:

1. **Stage 1 — Head Training:** Backbone weights were frozen; only the classification head was trained for initial convergence.

2. **Stage 2 — Fine-Tuning:** Backbone layers were gradually unfrozen and trained end-to-end with a reduced learning rate.

Training hyperparameters: batch size = 16; optimiser = Adam; callbacks = EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint. A fixed random seed of 42 was applied in Python, NumPy, and TensorFlow to ensure consistent and reproducible results across experiments.

6. Results and Discussion

This section presents the overall and class-wise results of the proposed model, along with a comparison to existing methods.

6.1 Overall Model Performance

Table 6 shows that fine-tuning in Stage 2 improved validation accuracy from 81.85% to 85.58% and reduced validation loss from 0.6298 to 0.5294.

Stage	Train Acc	Val Acc	Train Loss	Val Los
Stage 1 (Head Training)	0.7246	0.8185	0.7950	0.6298
Stage 2 (Fine-Tuned)	0.7628	0.8558	0.6890	0.5294

Figure 6. Overall model performance across training stages.

6.2 Class-wise Performance

Table 7 presents the per-class precision, recall, and F1-score on the test set. Performance varies substantially across classes, reflecting the inherent visual similarity between certain lesion types as well as residual class imbalance in the test set.

Class	Precision	Recall	F1-Score
Actinic Keratosis	0.6000	0.1875	0.2857
Basal Cell Carcinoma	0.5000	0.5625	0.5294
Dermatofibroma	0.6667	0.5000	0.5714
Melanoma	0.1429	0.0625	0.0870
Nevus	0.3429	0.7500	0.4706
Pigmented Benign Keratosis	0.4762	0.6250	0.5405
Seborrheic Keratosis	0.5800	0.3500	0.5600
Squamous Cell Carcinoma	0.3846	0.3125	0.3448
Vascular Lesion	0.6000	1.0000	0.7500

Figure 7. Class-wise precision, recall, and F1-score on the ISIC test set

Discussion of Class-wise Results

The model performed best on **Vascular Lesion** (F1 = 0.75, Recall = 1.00), indicating clear separability of its vascular patterns. **Dermatofibroma**, **Seborrheic Keratosis**, **Basal Cell**

Carcinoma, and **Pigmented Benign Keratosis** showed moderate performance, suggesting reasonable feature learning. In contrast, **Nevus** had high recall but low precision, indicating over-prediction. **Actinic Keratosis** showed low recall, meaning many true cases were missed. **Melanoma** remained the most challenging class, with the lowest precision, recall, and F1-score, mainly due to confusion with visually similar benign lesions..

6.3 Graphical Analysis

Training and Validation Curves

Figure 8 shows the accuracy and loss curves over both training stages. The validation accuracy improves monotonically in Stage 1 and continues to rise after fine-tuning in Stage 2, while the training loss decreases steadily, confirming effective learning without severe overfitting. The gap between training and validation accuracy narrows in Stage 2, indicating improved generalisation.

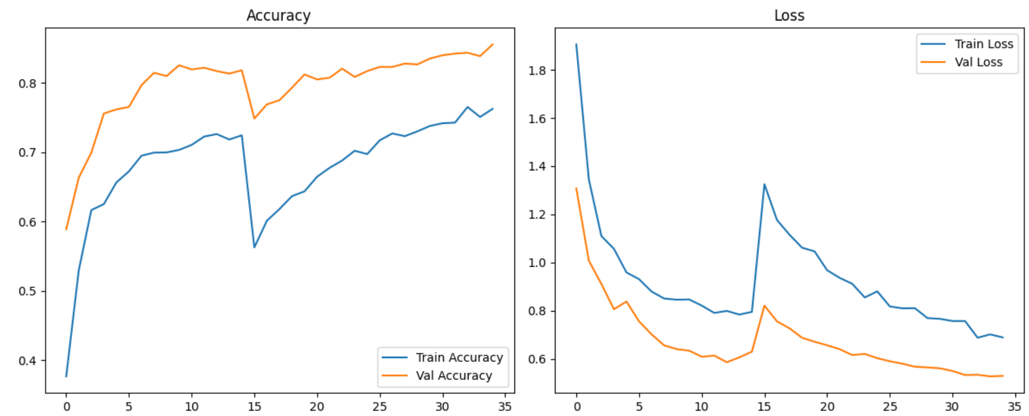


Figure 8. Training and validation accuracy (left) and loss (right) across both training stages.

Confusion Matrix

Figure 9 shows the 9×9 confusion matrix on the test set, where melanoma is often misclassified as nevus and actinic keratosis is confused across multiple classes, consistent with Table 7..

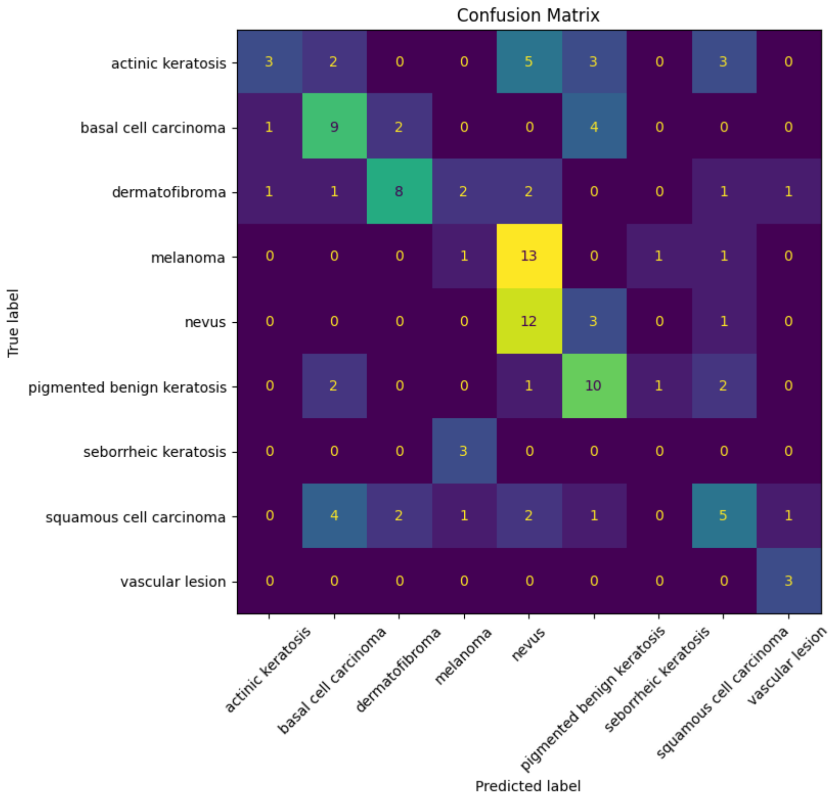


Figure 9. Confusion matrix on the ISIC test set (9 classes). Rows represent true labels; columns represent predicted labels.

6.4 Comparative Study

Table 10 compares the proposed approach against representative methods from the literature on the ISIC skin lesion dataset. The proposed method achieves competitive accuracy (85.58%) through the combination of domain-specific filter-based preprocessing and EfficientNetV2B0 fine-tuning, without relying on external segmentation masks or additional data sources.

Reference	Method	Backbone	Accuracy	F1-Score	Classes
Esteva et al. (2017)	CNN (End-to-End)	Inception V3	0.720	—	2
Haenssle et al. (2018)	Transfer Learning	ResNet-50	0.826	—	2
Tschandl et al. (2019)	EfficientNet	EfficientNetB4	0.847	0.710	7
Khan et al. (2021)	Hybrid CNN + SVM	VGG-16	0.831	0.780	7
Li et al. (2022)	Attention + ResNet	ResNet-101	0.860	0.740	9
Proposed	Filter Prep.+FT	EffNetV2B0	0.8558	0.4599	9

Figure 10. Comparative study of skin cancer detection methods on the ISIC dataset.

The proposed model shows an accuracy gain over basic transfer learning baselines and is competitive with more complex attention-based architectures, while using a significantly lighter backbone (EfficientNetV2B0). The lower macro F1 score relative to some references is partly attributable to the expanded 9-class task, which is inherently harder than 2- or 7-class formulations used in earlier work. Improving performance on the melanoma class remains the primary direction for future work.

7. Conclusion

This paper presented a multi-class skin lesion detection and classification framework that combines image preprocessing with deep learning. The proposed pipeline used CLAHE, median denoising, random oversampling, and a two-phase EfficientNetV2B0 transfer learning strategy to address image quality variation, class imbalance, and visual similarity between lesion classes.

The results showed that the framework is effective and computationally efficient for nine-class classification. Fine-tuning improved validation accuracy from 81.85% to 85.58%, while validation loss decreased from 0.6298 to 0.5294. Although the model performed well on several lesion classes, melanoma remained the most difficult category.

7.1 Limitations

The model was evaluated on a benchmark dataset and may not fully generalize to diverse real-world clinical settings. In addition, melanoma sensitivity remained low, and handcrafted texture features were not fully integrated into the final classifier.

7.2 Future Work

Future work will focus on improving melanoma detection, integrating hybrid feature fusion, and testing the framework on larger and more diverse clinical datasets. Explainable AI techniques may also be incorporated to improve model Accuracy transparency and clinical applicability. The development of real-time skin disease detection systems enabling dermatologists to perform quick and accurate diagnoses in real-time during patient consultations.It can greatly benefit telemedicine and mobile healthcare applications. Patients can capture images of their skin conditions using their smartphones, which are then processed by CNN models to provide preliminary diagnoses. This approach can improve access to dermatological care in remote areas and enable self monitoring for chronic skin conditions.

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